PROCEDURE

- Set up the sonometer on the table and clean the groove on the pulley to ensure minimum friction
- Stretch the wire by placing suitable mass in the hanger. Keep a small paper rider over the wire, between the two bridges.
- Set the tuning fork into vibrations by striking it against the rubber pad and place it over the sonometer,by its stem.
- Adjust the vibrating length of the wire by sliding the bridge B till the vibrating sound of the wire is maximum
- when the frequency of vibration is in resonance with the frequency of the tuning fork, the paper rider falls down.
- The length of the wire between the wedges A and B is measured using meter scale. It is called as resonant length.
- Repeat the above procedure for tuning forks of different frequencies by keeping the same load in the hanger.

OBSERVATIONS

Tension (constant) on the wire (mass suspended from the hanger including its own mass)
T = (mass suspended × 9.8) N

Variation of frequency with length		
Frequency of the tuning fork 'n ' (Hz)	Resonant length *'l'* $\times 10^{-2}$ m	nl
n_1 =		
n_2 =		
n_3 =		
n_4 =		
n_5 =		
n_6 =		

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